

All Data Structures Implementation in One Go –

1 - Linked List -

~1) LL Implementation

```
Programiz C++ Online Compiler
main.c... Output
25 void push_front(int val)
26 {
27     Node *newnode = new Node(val);
28     if (head == NULL)
29     {
30         head = tail = newnode;
31         return;
32     }
33     newnode->next = head;
34     head = newnode;
35 }
36 void push_back(int val)
37 {
38     Node *newnode = new Node(val);
39     if (head == NULL)
40     {
41         head = tail = newnode;
42         return;
43     }
44     tail->next = newnode;
45     tail = newnode;
46 }
47 void print_list()
48 {
49     Node *temp = head;
50     while (temp != NULL)
51     {
52         cout << temp->data << " ";
53         temp = temp->next;
54     }
55     cout << "NULL" << endl;
56 }
57 void insert_list(int val, int pos)
58 {
59     Node *newnode = new Node(val);
60     Node *temp = head;
61     for (int i = 0; i < pos - 1; i++)
62     {
63         if (temp == NULL)
64         {
65             cout << "Invalid position";
66             return;
67         }
68         temp = temp->next;
69     }
70     newnode->next = temp->next;
71     temp->next = newnode;
72 }
73 void pop_front()
74 {
75     if (head == NULL)
76     {
77         cout << "Nothing to pop";
78         return;
79     }
80     if (head == tail)
81     {
82         head = tail = NULL;
83         delete head;
84         return;
85     }
86     Node *temp = head;
87     head = head->next;
88     temp->next = NULL;
89     delete temp;
90 }
91 void pop_back()
92 {
93     if (head == NULL)
94     {
95         cout << "Invalid Tail";
96         return;
97     }
98     if (head == tail)
99     {
100         delete head;
101         head = tail = NULL;
102         return;
103     }
104     Node *temp = head;
105     while (temp->next->next != NULL)
106     {
107         temp = temp->next;
108     }
109     temp->next = NULL;
110     delete tail;
111     tail = temp;
112 }
113 };
114 int main()
115 {
116     List ll;
117 }
```

~1.1) DBLL Implementation

```
Programiz C++ Online Compiler
main.c... Output
4 class Node
5 {
6 public:
7     int data;
8     Node *next;
9     Node *prev;
10 Node(int val)
11 {
12     data = val;
13     next = prev = NULL;
14 }
15 };
16 class doublyList
17 {
18 public:
19     Node *head;
20     Node *tail;
21 doublyList()
22 {
23     head = tail = NULL;
24 }
25 void push_front(int val)
26 {
27     Node *newnode = new Node(val);
28     if (head == NULL)
29     {
30         head = tail = newnode;
31         return;
32     }
33     newnode->next = head;
34     head->prev = newnode;
35     head = newnode;
36 }
37 void push_back(int val)
38 {
39     Node *newnode = new Node(val);
40     if (tail == NULL)
41     {
42         head = tail = newnode;
43         return;
44     }
45     tail->next = newnode;
46     newnode->prev = tail;
47     tail = newnode;
48 }
49 void pop_front()
50 {
51     if (head == NULL)
52     {
53         return;
54     }
55     if (head == tail)
56     {
57         head = tail = NULL;
58         delete temp;
59         return;
60     }
61     head = head->next;
62     head->prev = NULL;
63     delete temp;
64 }
65 void pop_back()
66 {
67     if (tail == NULL)
68     {
69         return;
70     }
71     if (head == tail)
72     {
73         head = tail = NULL;
74         delete temp;
75         return;
76     }
77     tail = tail->prev;
78     tail->next = NULL;
79     delete temp;
80 }
81 void print_doublyList()
82 {
83     Node *temp = head;
84     while (temp != NULL)
85     {
86         cout << temp->data << "<->";
87         temp = temp->next;
88     }
89     cout << "NULL" << endl;
90 }
91 };
92 int main()
93 {
94     doublyList dll;
95     dll.push_front(2);
96     dll.push_front(4);
97     dll.push_front(6);
98     dll.print_doublyList();
99     dll.push_back(15);
100     dll.push_back(16);
101     dll.push_back(17);
102     dll.print_doublyList();
103     dll.pop_front();
104     dll.print_doublyList();
105     dll.pop_back();
106     dll.print_doublyList();
107 }
```

2 – Stack –

~2) Stack using LL Implementation

```

// Panel 1: Node and Stack classes
5 template <class T>
6 class Node
7 {
8 public:
9     T data;
10    Node *next;
11
12    Node(T val)
13    {
14        data = val;
15        next = NULL;
16    };
17 };
18
19 template <class T>
20 class Stack
21 {
22     Node<T> *head;
23 public:
24     Stack()
25     {
26         head = NULL;
27     }
28
29     void push_front(T val)
30     {
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```

```

4 // Queue Implemented by LL
5 class Node
6 {
7 public:
8     int data;
9     Node *next;
10    Node(int data)
11    {
12        this->data = data;
13        this->next = NULL;
14    }
15 };
16
17 class Queue
18 {
19     Node *head;
20     Node *tail;
21 public:
22     Queue()
23     {
24         head = NULL;
25         tail = NULL;
26     }
27
28     void push(int data)
29     {
30
31         Node *newnode = new Node(data);
32         if (head == NULL)
33         {
34             head = tail = newnode;
35         }
36         else
37         {
38             tail->next = newnode;
39             tail = newnode;
40         }
41     }
42
43     int pop()
44     {
45         if (isEmpty())
46         {
47             cout << "Queue is already empty";
48             return -1;
49         }
50         Node *temp = head;
51         head = head->next;
52         delete temp;
53     }
54
55     int front()
56     {
57         if (isEmpty())
58         {
59             cout << "Already Empty";
60             return -1;
61         }
62         return head->data;
63     }
64
65     bool isEmpty()
66     {
67         return head == NULL;
68     }
69 };
70
71 int main()
72 {
73     Queue q;
74     q.push(1);
75     q.push(2);
76     q.push(3);
77     q.push(4);
78
79     cout << "Elements of the Queue are - " << endl;
80     while (!q.isEmpty())
81     {
82         cout << q.front() << " ";
83         q.pop();
84     }
85     cout << endl;
86 }
87
88 /*
89 Elements of the Queue are -
90 1
91 2
92 3
93 4
94 */

```

~3.1) Queue using vector Implementation

```

4 // Queue Implemented by vector
5 class Queue
6 {
7     vector<int> vec;
8 public:
9     void push(int val)
10    {
11        vec.push_back(val);
12    }
13
14    bool isEmpty()
15    {
16        return vec.size() == 0;
17    }
18
19    void pop()
20    {
21        if (isEmpty())
22        {
23            cout << "Queue is empty, nothing to pop" << endl;
24            return;
25        }
26        vec.erase(vec.begin());
27    }
28
29    int front()
30    {
31        if (isEmpty())
32        {
33            cout << "Queue is empty" << endl;
34            return -1;
35        }
36        return vec[0];
37    }
38
39    int main()
40    {
41        Queue q;
42        q.push(1);
43        q.push(2);
44        q.push(3);
45        q.push(4);
46
47        cout << "Elements of the Queue are - " << endl;
48        while (!q.isEmpty())
49        {
50            cout << q.front() << " ";
51            q.pop();
52        }
53        cout << endl;
54    }
55 }
56
57 /*
58 Queue is -
59 1 2 3 4
60 */

```

~3.2) Circular Queue Implementation

```

4 class CircularQueue
5 {
6     int *arr;
7     int currSize, n;
8     int f, r;
9
10 public:
11     CircularQueue(int size)
12     {
13         n = size;
14         arr = new int[n];
15         currSize = 0;
16         f = 0;
17         r = -1;
18     }
19
20     void push(int data)
21     {
22         if (currSize == n)
23         {
24             cout << "CQ is FULL" << endl;
25             return;
26         }
27         r = (r + 1) % n;
28         arr[r] = data;
29         currSize++;
30     }
31
32     void pop()
33     {
34         if (empty())
35         {
36             cout << "CQ is Already Empty " << endl;
37             return;
38         }
39         f = (f + 1) % n;
40         currSize--;
41     }
42
43     int front()
44     {
45         if (empty())
46         {
47             cout << "CQ is Already Empty " << endl;
48             return 0;
49         }
50         return arr[f];
51     }
52
53     bool empty()
54     {
55         return currSize == 0;
56     }
57
58     void printArray()
59     {
60         for (int i = 0; i < n; i++)
61         {
62             cout << arr[i] << " ";
63         }
64         cout << endl;
65     }
66 };
67
68 int main()
69 {
70     CircularQueue cq(3);
71     cq.push(1);
72     cq.push(2);
73     cq.push(3);
74     cq.pop();
75     cq.push(4);
76     cq.printArray(); // 4 2 3
77     cout << endl;
78     while (!cq.empty())
79     {
80         cout << cq.front() << " ";
81         cq.pop();
82     }
83     cout << endl; // 2 3 4
84 }

```

4 – Binary Tree –

~4) Binary Tree Implementation

```

4 class Node
5 {
6 public:
7     int data;
8     Node *left;
9     Node *right;
10
11     Node(int val)
12     {
13         data = val;
14         left = right = NULL;
15     }
16 };
17
18 static int idx = -1;
19
20 Node *createTree(vector<int> nodes)
21 {
22     idx++;
23     if (nodes[idx] == -1)
24     {
25         return NULL;
26     }
27     Node *currNode = new Node(nodes[idx]);
28     currNode->left = createTree(nodes);
29     currNode->right = createTree(nodes);
30     return currNode;
31 }
32
33 void preOrder(Node *root)
34 {
35     if (root == NULL)
36     {
37         return;
38     }
39     cout << root->data << " ";
40     preOrder(root->left);
41     preOrder(root->right);
42 }
43
44 void inOrder(Node *root)
45 {
46     if (root == NULL)
47     {
48         return;
49     }
50     inOrder(root->left);
51     cout << root->data << " ";
52     inOrder(root->right);
53 }
54
55 void postOrder(Node *root)
56 {
57     if (root == NULL)
58     {
59         return;
60     }
61     postOrder(root->left);
62     postOrder(root->right);
63     cout << root->data << " ";
64 }
65
66 void levelOrder(Node *root)
67 {
68     if (root == NULL)
69     {
70         return;
71     }
72     queue<Node*> q;
73     q.push(root);
74
75     while (!q.empty())
76     {
77         Node *curr = q.front();
78         q.pop();
79         cout << curr->data << " ";
80         if (curr->left != NULL)
81             q.push(curr->left);
82         if (curr->right != NULL)
83             q.push(curr->right);
84     }
85 }
86
87 void modifiedLevelOrderTraversal(Node *root)
88 {
89     if (root == NULL)
90     {
91         return;
92     }
93     queue<Node*> q;
94     q.push(root);
95     q.push(NULL);
96
97     while (!q.empty())
98     {
99         Node *curr = q.front();
100        q.pop();
101        if (curr->left != NULL)
102            q.push(curr->left);
103        if (curr->right != NULL)
104            q.push(curr->right);
105    }
106 }

```

~4.1) Binary Tree All Functions

```

107     if (curr == NULL)
108     {
109         cout << " " << endl;
110         if (q.empty())
111         {
112             break;
113         }
114         q.push(NULL);
115     }
116     else
117     {
118         cout << curr->data << " ";
119     }
120 }
121
122 if (curr->left != NULL)
123 {
124     q.push(curr->left);
125 }
126 if (curr->right != NULL)
127 {
128     q.push(curr->right);
129 }
130 }
131 }
132
133 int heightofTree(Node *root)
134 {
135     if (root == NULL)
136     {
137         return 0;
138     }
139     int leftHeight = heightofTree(root->left);
140     int rightHeight = heightofTree(root->right);
141     int currHeight = max(leftHeight, rightHeight) + 1;
142     return currHeight;
143 }
144
145 int countofNodes(Node *root)
146 {
147     if (root == NULL)
148     {
149         return 0;
150     }
151     int leftCount = countofNodes(root->left);
152     int rightCount = countofNodes(root->right);
153     int finalCount = leftCount + rightCount + 1;
154     return finalCount;
155 }
156
157 int sumofNodes(Node *root)
158 {
159     if (root == NULL)
160     {
161         return 0;
162     }
163     int leftSum = sumofNodes(root->left);
164     int rightSum = sumofNodes(root->right);
165     int totalSum = leftSum + rightSum + root->data;
166     return totalSum;
167 }
168
169 int main()
170 {
171     vector<int> nodes = {1, 2, 4, -1, -1, 5, -1, -1, 3, -1, 6, -1, -1};
172     Node *root = createTree(nodes);
173     cout << "root is - " << root->data << endl;
174     cout << "Preorde is - " << endl;
175     preOrder(root);
176     cout << "Inorder Traversal is - " << endl;
177     inOrder(root);
178     cout << "PostOrder is - " << endl;
179     postOrder(root);
180     cout << "Levelorder Traversal is - " << endl;

```

5 – BST –

~5) BST Implementation

```

4 class Node
5 {
6 public:
7     int data;
8     Node *left;
9     Node *right;
10    Node(int val)
11    {
12        data = val;
13        left = right = NULL;
14    }
15 };
16
17 Node *insert(Node *root, int val)
18 {
19     if (root == NULL)
20     {
21         return new Node(val);
22     }
23     if (root->data == val)
24     {
25         return root;
26     }
27     if (root->data > val)
28     {
29         root->left = insert(root->left, val);
30     }
31     else
32     {
33         root->right = insert(root->right, val);
34     }
35     return root;
36 }
37
38 Node *createBST(vector<int> nodes, int n)
39 {
40     Node *root = NULL;
41     for (int i = 0; i < n; i++)
42     {
43         root = insert(root, nodes[i]);
44     }
45     return root;
46 }
47
48 bool searchBST(Node *root, int k)
49 {
50     if (root == NULL)
51         return false;
52     if (root->data == k)
53         return true;
54     else if (root->data > k)
55         return searchBST(root->left, k);
56     else
57         return searchBST(root->right, k);
58 }
59
60 Node *inorderSuccessor(Node *root)
61 {
62     while (root->left != NULL)
63         root = root->left;
64     return root;
65 }
66
67 Node *deleteBST(Node *root, int todelete)
68 {
69     if (root == NULL)
70         return NULL;
71     if (root->data > todelete)
72     {
73         root->left = deleteBST(root->left, todelete);
74     }
75     else if (root->data < todelete)
76     {
77         root->right = deleteBST(root->right, todelete);
78     }
79 }

```

~5.1) BST Operations

```

79 else
80 {
81     if (root->left == NULL && root->right == NULL)
82     {
83         delete root;
84         return NULL;
85     }
86     else if (root->left == NULL || root->right == NULL)
87     {
88         Node *child;
89         if (root->left != NULL)
90             child = root->left;
91         else
92             child = root->right;
93         delete root;
94         return child;
95     }
96     Node *is = inorderSuccessor(root->right);
97     root->data = is->data;
98     root->right = deleteBST(root->right, is->data);
99 }
100 return root;

```

```

101 return root;
102 }
103
104 void inorder(Node *root)
105 {
106     if (root == NULL)
107         return;
108     inorder(root->left);
109     cout << root->data << " ";
110     inorder(root->right);
111 }
112
113 int main()
114 {
115     vector<int> nodes = {5, 1, 3, 4, 2, 7, 9};
116     int n = nodes.size();
117     Node *root = createBST(nodes, n);
118     cout << "Created BST is -" << endl;
119     inorder(root);
120     cout << endl;
121     cout << searchBST(root, 4) << endl;
122     cout << searchBST(root, 15) << endl;
123     cout << searchBST(root, 9) << endl;
124     cout << searchBST(root, 0) << endl;
125 }

```

```

113 vector<int> nodes = {5, 1, 3, 4, 2, 7, 9};
114 int n = nodes.size();
115 Node *root = createBST(nodes, n);
116 cout << "Created BST is -" << endl;
117 inorder(root);
118 cout << endl;
119 cout << searchBST(root, 4) << endl;
120 cout << searchBST(root, 15) << endl;
121 cout << searchBST(root, 9) << endl;
122 cout << searchBST(root, 0) << endl;
123
124 root = deleteBST(root, 4);
125 inorder(root);
126 cout << endl;
127 root = deleteBST(root, 15);
128 inorder(root);
129 cout << endl;
130 root = deleteBST(root, 9);
131 inorder(root);
132 cout << endl;
133 }

```

6 – Heap –

~6) Heap Sort Implementation

```

3 // Max Heap Implementation -
4 class Heap
5 {
6     vector<int> vec;
7
8 public:
9     void push(int val)
10    {
11        vec.push_back(val);
12        int childIdx = vec.size() - 1;
13        int parIdx = (childIdx - 1) / 2;
14        while (childIdx > 0 && vec[childIdx] > vec[parIdx])
15        {
16            swap(vec[childIdx], vec[parIdx]);
17            childIdx = parIdx;
18            parIdx = (childIdx - 1) / 2;
19        }
20    }
21
22    void heapify(int i)
23    {
24        if (i >= vec.size()) return;
25        int l = 2 * i + 1;
26        int r = 2 * i + 2;

```

```

28     int r = 2 * i + 2;
29     int maxIdx = i;
30     if (l < vec.size() && vec[l] > vec[maxIdx]) maxIdx = l;
31     if (r < vec.size() && vec[r] > vec[maxIdx]) maxIdx = r;
32     if (maxIdx != i)
33     {
34         swap(vec[i], vec[maxIdx]);
35         heapify(maxIdx);
36     }
37
38 void pop()
39 {
40     if (vec.size() == 0) return;
41     swap(vec[0], vec[vec.size() - 1]);
42     vec.pop_back();
43     heapify(0);
44 }
45
46 int top()
47 {
48     return vec[0];
49 }

```

```

50 int top()
51 {
52     if (vec.size() == 0) return -1;
53     return vec[0];
54 }
55
56 bool isempty()
57 {
58     return vec.size() == 0;
59 }
60
61 int main()
62 {
63     Heap heap;
64     heap.push(20);
65     heap.push(150);
66     heap.push(70);
67     heap.push(100);
68     while (!heap.isempty())
69     {
70         cout << heap.top() << " ";
71         heap.pop();
72     }
73 }

```